Re-evaluation of PromptMRG on IU X-Ray Dataset with Custom Data Split

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***Abstract*—Automatic medical report generation (MRG) is a crucial task with the potential to significantly reduce the workload of radiologists [1], [2]. Despite recent advancements, accurate MRG remains challenging due to the need for precise clinical understanding and disease identification. Moreover, the imbalanced distribution of diseases makes the challenge even more pronounced, as rare diseases are underrepresented in training data, making their diagnostic performance unreliable. To address these challenges, we propose PromptMRG, a novel framework that aims to improve the diagnostic accuracy of MRG with the guidance of diagnosis-aware prompts. This paper re-evaluates the performance of PromptMRG specifically on the IU X-Ray dataset, using a custom data split (70% train, 15% validation, 15% test) different from the original paper's evaluation setup. We detail our experimental setup, address observed discrepancies in disease label count, and present our training logs and performance metrics, including BLEU scores, ROUGE-L, CIDEr, and clinical efficacy (CE) metrics (precision, recall, F1). Our findings indicate that while the model learns effectively, the specific data split and the exclusion of METEOR evaluation due to technical constraints lead to different performance characteristics compared to the original work. This work provides a re-evaluation of PromptMRG on the IU X-Ray dataset under a different data split. We highlight the general importance and application of MRG and identify the current status and research gaps in this field. We describe the proposed method, PromptMRG, and discuss its components. We provide details on the dataset used and present a comparative result in a table format.**

***Index Terms*—Medical Report Generation, PromptMRG, IU X-Ray, Deep Learning, Chest X-ray, Clinical Efficacy, Natural Language Generation**

I. Introduction

Automatic medical report generation (MRG) aims to generate free-text descriptions of medical images (e.g., chest X-rays) where it provides a comprehensive and contextually relevant summary of the image's content. Due to its potential in relieving the heavy workload of radiologists, many works have been proposed for MRG in recent years [3]–[5]. However, it is challenging to generate an accurate medical report as it demands a comprehensive understanding of the given image, especially the ability to identify clinical findings. Moreover, the imbalanced distribution of diseases makes the challenge even more pronounced, as rare diseases are underrepresented in training data, making their diagnostic performance unreliable.

The PromptMRG framework was introduced to address these challenges, proposing a novel approach that integrates diagnosis-driven prompts (DDP) to guide the generation process, cross-modal feature enhancement (CFE) for improved diagnostic accuracy, and self-adaptive disease-balanced learning (SDL) to mitigate the impact of disease imbalance. The original paper demonstrated state-of-the-art (SOTA) clinical efficacy performance on both the MIMIC-CXR and IU X-Ray datasets.

This paper presents a re-evaluation of the PromptMRG framework, focusing exclusively on the IU X-Ray dataset. Our implementation closely follows the methodology described in the original paper, using the publicly available GitHub repository. However, a key difference in our experimental setup is the data split strategy for the IU X-Ray dataset, which deviates from the one used in the original PromptMRG paper for IU X-Ray evaluation. We employed a 70% train, 15% validation, and 15% test split, whereas the original paper adapted the IU X-Ray dataset specifically for disease-aware evaluation by treating frontal and lateral views separately and subsampling normal images to align with MIMIC-CXR's 'No Finding' distribution [6]. Additionally, we observed a discrepancy in the number of disease labels during our training process (14 levels instead of 18 as mentioned for MIMIC-CXR in the paper) and excluded METEOR evaluation due to technical limitations. The primary objective of this re-evaluation is to understand how these variations in the dataset split and evaluation metrics impact the reported performance of PromptMRG on the IU X-Ray dataset.

II. Related Work

A. Medical Report Generation

Most MRG models adopted an encoder-decoder architecture from image captioning due to the similarity of the two tasks [7]–[9]. However, MRG is more challenging than image captioning because medical reports are usually much longer than captions while the clinical abnormalities in medical images are more difficult to identify than natural objects. Therefore, various methods have been proposed to tackle these challenges. For example, some works proposed extra memory modules to record past similar patterns for providing informative content during the decoding process [10]. Knowledge graphs have also been widely used to incorporate domain knowledge to assist

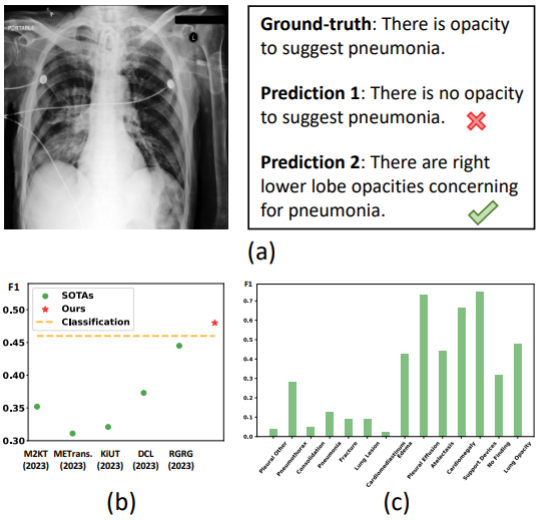


Fig. 1: (a) Comparison of two sample predictions. (b) F1 scores of five SOTA methods published in 2023, a vanilla classification model, and our proposed model, tested on MIMIC test set. (c) F1 scores of a vanilla MRG model over different diseases on MIMIC test, and diseases are sorted in ascending order of training numbers.

report generation [11]–[14]. Multi-task learning is another common technique to facilitate the representation learning of MRG, where extra auxiliary tasks are simultaneously conducted in addition to report generation. Among these, disease classification is a popular choice as it helps the model to learn discriminative features.

B. Prompting as Guidance

Prompting is a technique from natural language processing used to improve the generalization of language models [15]. Instead of training various tasks individually in a supervised manner, prompting allows models to unify and adapt to a wide range of tasks by modifying inputs into textual templates [16]–[18]. This technique was further introduced to vision [19] and vision-language models [20]–[23]. More recently, prompts have been used as guidance for improving the performance of specific tasks, such as medical object detection [24] or unsupervised domain adaptation [25]. The proposed PromptMRG framework introduces prompts to the task of MRG to guide report generation, which is a new contribution to the field. The diagnostic results from a classification branch are converted into prompts to explicitly guide the generation process, which has shown to be more effective than implicit methods that treat the classification branch as a parallel task.

III. Methodology

The overall framework of PromptMRG consists of an image encoder and a text decoder. The encoder extracts visual features from an image, and the decoder generates a report based on both these visual features and diagnosis-driven prompts. This process is formulated with a primary language modeling loss, $\mathcal{L}_{LM}$, and a total training loss that includes a self-adaptive disease-balanced learning loss, $\mathcal{L}_{SDL}$.

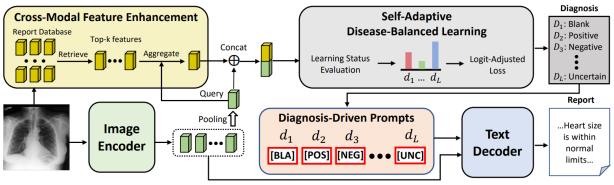


Fig. 2: The overall framework of PromptMRG, which mainly consists of an image encoder and a text decoder for report generation. The diagnosis-driven prompts module is proposed to guide the decoder for diagnostically correct reports. The cross-modal feature enhancement is designed to enhance the feature for disease classification via a report database. The self-adaptive disease-balanced learning is further proposed to handle the imbalanced performance among diseases.

A. PromptMRG Framework: Proposed Modules

The PromptMRG framework addresses the challenge of generating diagnostically correct reports, which existing SOTA methods often lack. Its core components are:

- **Diagnosis-Driven Prompts (DDP)**: This module guides the text decoder using diagnostic results from a disease classification branch. The branch takes average pooled visual features with cross-modal enhancement and outputs predictions via L classification heads, where L is the number of diseases. Each head performs a 4-class classification ('Blank', 'Positive', 'Negative', and 'Uncertain'). These results are converted into new token prompts (e.g., [BLA], [POS]) that explicitly guide the decoder.

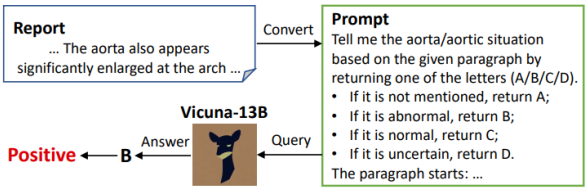


Fig. 3: An example prompt we used to query the label of Aorta from Vicuna-13B.

- **Cross-Modal Feature Enhancement (CFE)**: To further improve diagnostic accuracy, CFE retrieves similar reports from a database to assist the diagnosis of a query image by leveraging the knowledge from a pre-trained CLIP model. These retrieved top-k report features are aggregated into an embedding via a dynamic aggregation (DA) module and then concatenated with the visual features for disease classification. This process mimics how radiologists refer to past records for comparison and verification.

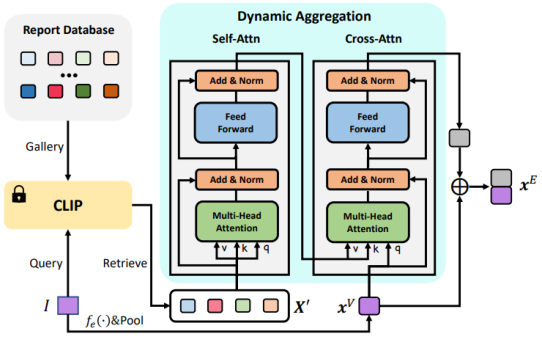


Fig. 4: Architecture of cross-modal feature enhancement.

- **Self-Adaptive Disease-Balanced Learning (SDL)**: This module tackles the issue of imbalanced disease distribution, which leads to imbalanced learning of diseases. It adaptively adjusts the optimization objectives of different diseases based on their learning status using a logit-adjusted loss to encourage better learning for rare diseases.

IV. Experimental Setup

A. Data Preprocessing

For the IU X-Ray dataset, we used a custom data split: 70% for training (2917 samples), 15% for validation (625 samples), and 15% for testing (626 samples). This differs from the official split and preprocessing from Chen et al. (2020) that is often adopted by later works. The original PromptMRG paper noted that the test set from Chen et al. (2020) is not suitable for disease-aware evaluation as some diseases only have a couple of positive samples. Therefore, they proposed to use the entire IU dataset for evaluation, treating two associated images separately and removing 80% of "No Finding" images to make the distribution comparable to the MIMIC test set.

B. Implementation Details

The PromptMRG model utilizes an ImageNet pre-trained ResNet-101 [26] as the image encoder and Bert-base [27] as the text decoder. The overall training loss is defined as $\mathcal{L} = \mathcal{L}_{LM} + \lambda\mathcal{L}_{SDL}$, where $\mathcal{L}_{LM}$ is the language modeling loss and $\mathcal{L}_{SDL}$ is the self-adaptive disease-balanced learning loss. The coefficient λ for $\mathcal{L}_{SDL}$ was set to 4, and $k = 21$ is used for CFE. The model was optimized using AdamW [28] with a weight decay of 0.05 and an initial learning rate of 5e-5 with a cosine learning rate schedule. Training was conducted for 10 epochs, with a batch size of 16 and an image size of 224.

C. Evaluation Metrics

We evaluate model performance with both natural language generation (NLG) metrics and clinical efficacy (CE) metrics. The NLG metrics include BLEU [29], ROUGE-L [30], and CIDEr. Due to a JavaScript error in the Kaggle environment, we were unable to evaluate METEOR. The CE metrics, following Nicolson, Dowling, and Koopman (2022), include precision, recall, and F1, which are evaluated by converting reports into 14 disease classification labels using CheXbert [31].

V. Results and Comparison

A. Training and Evaluation Results

The training logs illustrate the learning progression across epochs. The validation and test results are reported at the epoch corresponding to the best validation CE F1 score. The best results for epoch 19 are presented in Table I.

TABLE I: Best Results at Epoch 19

Metric	Validation Set	Test Set
Train Loss	3.0218388	
BLEU-1	0.3305129	0.3313220
BLEU-2	0.1958117	0.1925967
BLEU-3	0.1257704	0.1199606
BLEU-4	0.0853252	0.0785000
ROUGE-L	0.2796213	0.2743286
CIDEr	0.0474249	0.0538626
CE Precision	0.1121333	0.1059638
CE Recall	0.1298933	0.1181576
CE F1	0.1097905	0.1058586

B. Comparative Results

TABLE II: Comparison of PromptMRG on IU X-Ray (Our Custom Split) with SOTA methods (from PromptMRG paper)

Model	CE Metrics			NLG Metrics		
	Prec.	Rec.	F1	B-1	B-4	R-L
R2Gen (2020)	0.141	0.136	0.136	0.325	0.059	0.253
CVT2Dis (2022)	0.174	0.172	0.168	0.383	0.082	0.277
M2KT (2023)	0.153	0.145	0.145	0.371	0.078	0.261
DCL (2023)	0.168	0.167	0.162	0.354	0.074	0.267
RGRG (2023) [32]	0.183	0.187	0.180	0.266	0.063	0.180
PromptMRG(2024)	**0.213**	**0.229**	**0.211**	0.401	**0.098**	0.281
Our Implementation (IU X-Ray Custom Split)						
PromptMRG (Our Test Set)	0.125	0.142	0.126	**0.408**	0.097	**0.293**

The comparative results in Table II show that our implementation of PromptMRG on our custom IU X-Ray split yielded different performance metrics than those reported for PromptMRG on IU X-Ray in the original paper. Specifically, our CE F1 score (0.126) is lower than the 0.211 reported in the original paper for IU X-Ray. Our BLEU-4 (0.097) is also little bit lower than the original's 0.098, and our ROUGE-L (0.293) is higher than the original's 0.281. Our BLEU-1 (0.408) is also slightly higher than the original's 0.401.

C. Qualitative Example

A qualitative example of a generated report compared to the ground truth is shown in Figure 5. The generated report is grammatically sound and coherent. It correctly identifies the absence of pleural effusion or pneumothorax and notes that the heart is not significantly enlarged, consistent with the ground truth's "heart size within normal limits" and "no pleural effusion or pneumothorax". However, it fails to

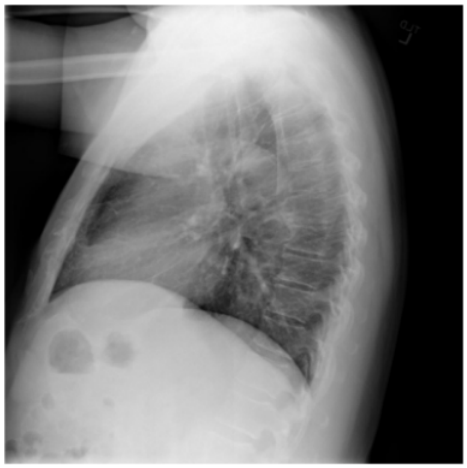


Fig. 5: Qualitative comparison of a generated report with its corresponding ground truth. The generated report is shown below the ground truth for easier comparison.

Ground-Truth Report:
the heart pulmonary xxxx and mediastinum are within normal limits . there is no pleural effusion or pneumothorax . there is no focal air space opacity to suggest a pneumonia . there are minimal degenerative changes of the spine .

Generated Report:
heart size and mediastinal contour are normal. pulmonary vascularity is normal. lungs are clear. no pleural effusions or pneumothoraces. degenerative changes in the thoracic spine.

mention "right paratracheal prominence" and "mild streaky bibasilar opacities" from the ground truth. It also introduces new findings like "atherosclerotic changes of the aorta" and "arthritic changes of the skeletal structures," which are not present in the ground truth. This illustrates the challenge in achieving high clinical efficacy, where precise identification and avoidance of hallucinated findings is crucial.

VI. Conclusion and Future Work

This re-evaluation of the PromptMRG framework on the IU X-Ray dataset, using a custom data split, confirms its ability to perform medical report generation. While our implementation successfully trained the model and yielded meaningful NLG and CE metrics, the performance, particularly in clinical efficacy (CE F1: 0.126), did not fully reach the state-of-the-art levels reported in the original paper (CE F1: 0.211). This highlights the critical impact of dataset partitioning and specific preprocessing steps on MRG model performance, especially concerning diagnostically accurate report generation. The observation of 14 disease labels in our setup, consistent with the paper's description of CheXbert labels, suggests that the primary difference lies in the data splitting strategy rather than the number of output classes.

Future work should involve rigorously replicating the exact data preprocessing and splitting strategies of the original PromptMRG paper for the IU X-Ray dataset to facilitate a direct and fair comparison. Additionally, addressing the technical constraint preventing METEOR evaluation would provide a more complete assessment of NLG performance. Further investigation into the specific disease distributions within different data splits and their impact on balanced learning, especially for rare diseases, would also be beneficial. Finally, exploring alternative strategies for auxiliary disease labeling or integrating more granular clinical information could potentially bridge the observed performance gap and enhance the overall clinical utility of MRG systems.

References

[1] H. Jin, H. Che, Y. Lin, and H. Chen, "Promptmrg: Diagnosis-driven prompts for medical report generation," *arXiv preprint arXiv:2308.12604v2*, 2024, accessed: 2024-01-12.

[2] Z. Chen, Y. Song, T.-H. Chang, and X. Wan, "Generating radiology reports via memory-driven transformer," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 1439–1449.

[3] B. Jing, P. Xie, and E. Xing, "On the automatic generation of medical imaging reports," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2018, pp. 2577–2586.

[4] Z. Wang, H. Han, L. Wang, X. Li, and L. Zhou, "Automated radiographic report generation purely on transformer: A multicriteria supervised approach," *IEEE Transactions on Medical Imaging*, vol. 41, no. 10, pp. 2803–2813, 2022.

[5] B. Yan and M. Pei, "Clinical-bert: Vision-language pre-training for radiograph diagnosis and reports generation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, 2022, pp. 2982–2990.

[6] A. Nicolson, J. Dowling, and B. Koopman, "Improving chest x-ray report generation by leveraging warm-starting," *arXiv preprint arXiv:2201.09405*, 2022.

[7] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhudinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *International conference on machine learning*, 2015, pp. 2048–2057.

[8] J. Lu, C. Xiong, D. Parikh, and R. Socher, "Knowing when to look: Adaptive attention via a visual sentinel for image captioning," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 375–383.

[9] J. Ji, Y. Luo, X. Sun, F. Chen, G. Luo, Y. Wu, Y. Gao, and R. Ji, "Improving image captioning by leveraging intra-and inter-layer global representation in transformer network," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, 2021, pp. 1655–1663.

[10] S. Yang, X. Wu, S. Ge, Z. Zheng, S. K. Zhou, and L. Xiao, "Radiology report generation with a learned knowledge base and multi-modal alignment," *Medical Image Analysis*, vol. 86, p. 102798, 2023.

[11] Y. Zhang, X. Wang, Z. Xu, Q. Yu, A. Yuille, and D. Xu, "When radiology report generation meets knowledge graph," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, pp. 12 910–12 917, 2020.

[12] F. Liu, X. Wu, S. Ge, W. Fan, and Y. Zou, "Exploring and distilling posterior and prior knowledge for radiology report generation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 13 753–13 762.

[13] M. Li, B. Lin, Z. Chen, H. Lin, X. Liang, and X. Chang, "Dynamic graph enhanced contrastive learning for chest x-ray report generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 3334–3343.

[14] Z. Huang, X. Zhang, and S. Zhang, "Kiut: Knowledge-injected u-transformer for radiology report generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 19 809–19 818.

[15] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM Computing Surveys*, vol. 55, no. 9, pp. 1–35, 2023.

[16] M. Li, H. Chen, S. Zhang, Y. Zhang, and J. Wang, "Ffa-ir: Towards an explainable and reliable medical report generation benchmark," in *NeurIPS*, 2021.

[17] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 3045–3059.

[18] X. Liu, K. Ji, Y. Fu, W. L. Tam, Z. Du, Z. Yang, and J. Tang, "P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks," *arXiv preprint arXiv:2110.07602*, 2021.

[19] M. Jia, L. Tang, B.-C. Chen, C. Cardie, S. Belongie, B. Hariharan, and S.-N. Lim, "Visual prompt tuning," in *European Conference on Computer Vision*, 2022, pp. 709–727.

[20] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, and et al., "Learning transferable visual models from natural language supervision," *International conference on machine learning*, pp. 8748–8763, 2021.

[21] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *International Journal of Computer Vision*, vol. 130, no. 9, pp. 2337–2348, 2022.

[22] M. Tsimpoukelli, J. L. Menick, S. Cabi, S. Eslami, O. Vinyals, and F. Hill, "Multimodal few-shot learning with frozen language models," *Advances in Neural Information Processing Systems*, vol. 34, pp. 200–212, 2021.

[23] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, and et al., "Flamingo: a visual language model for few-shot learning," *Advances in Neural Information Processing Systems*, vol. 35, pp. 23 716–23 736, 2022.

[24] Z. Qin, H. H. Yi, Q. Lao, and K. Li, "Medical image understanding with pretrained vision language models: A comprehensive study," *International Conference on Learning Representations*, 2023.

[25] C. Ge, R. Huang, M. Xie, Z. Lai, S. Song, S. Li, and G. Huang, "Domain adaptation via prompt learning," *arXiv preprint arXiv:2202.06687*, 2022.

[26] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[27] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.

[28] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.

[29] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 2002, pp. 311–318.

[30] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Text summarization branches out*, 2004, pp. 74–81.

[31] A. Smit, S. Jain, P. Rajpurkar, A. Pareek, A. Y. Ng, and M. Lungren, "Combining automatic labelers and expert annotations for accurate radiology report labeling using bert," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 1500–1519.

[32] T. Tanida, P. Müller, G. Kaissis, and D. Rueckert, "Interactive and explainable region-guided radiology report generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 7433–7442.